

Evolutionary dynamics of language: Insights from a diverse sample of sociolinguistic studies

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Abstract

The evolutionary dynamics of language can be studied using phylogenetic modelling of changes over thousands of years, or by close observation of changes unfolding over a few decades at the community level. While the phylogenetic approach uses data from hundreds of languages to make cross-linguistic generalisations, community-level studies of sociolinguistic variation have until recently been limited to a very narrow language sample. However this is now changing, with an increasing number of variationist studies published on diverse languages. In this article I assemble a diverse sample of variationist studies encompassing 63 languages from 26 families, and explore potential patterns regarding rates of change in linguistic features, and which features are associated with social categories or social identity signalling. These observations largely converge with results obtained from phylogenetic methods, suggesting that more systematic meta-analysis of diverse variationist studies will provide a new way to investigate the evolutionary dynamics of language.

1. Introduction

Linguistics has taken an evolutionary turn in the last two decades, with new methods connecting the close study of languages with broader questions about evolutionary dynamics. Evolutionary linguists on the one hand investigate the origins of human language (which will not be further considered here), and on the other hand processes of language change and diversification. The latter concerns overlap with both historical linguistics and sociolinguistics, but evolutionary linguists use a different range of tools, most notably Bayesian techniques for estimating the likelihood of historical scenarios, i.e. phylogenetic modelling. These historical inferences include estimated rates at which linguistic features change, and which changes enhance diversification among related languages (Gray et al. 2013; Goldstein 2022). Another key concern, linked to evolutionary theory, is the social and ecological factors that influence rates of change (Greenhill 2015).

Evolutionary linguists also must zoom in and out between different timeframes (Enfield 2014: 9). Alongside big-picture phylogenies spanning thousands of years, we also need to understand the socio-cultural processes by which innovations emerge and potentially spread through communities (Evans 2018; Roberts & Sneller 2020). This can inform analyses of rates of change, since social factors could either retard or accelerate changes. For example, smaller populations may allow more rapid changes, because a new variant could more easily spread to all members of the community (Nettle 1999a; Nettle 1999b), though empirical support for this remains unclear (Bromham 2024). Another possibility is that social group identity can fuel language change, if speakers actively prefer innovative forms to reflect social distinctions. It is on this point that sociolinguistics makes its most important contribution to evolutionary linguistics. We can formulate the following hypothesis: *whichever linguistic features are more likely to signal social identity, should on a larger scale show more rapid rates of change.* To the extent that findings from sociolinguistics agree with phylogenetic estimates of rates of change, this could produce a valuable triangulation of research methods, with convergent findings providing stronger support than either method on its own (Sinnemäki 2014).

Sociolinguistic studies of language change gained new impetus in the 1960s, using variation within a community as a window into processes of change (Weinreich et al. 1968).

Variationist sociolinguists investigate whether linguistic variants (e.g. *isn't* ~ *ain't*) are statistically associated with social factors such as geography, gender, age or social class. They also sometimes report whether speakers find variants to be noticeable or socially meaningful. Thus sociolinguistics can shed light on social drivers of language change. But such findings only become truly informative for broader evolutionary theory if they generalise over a range of languages and social settings, and it is only recently that variationist sociolinguistics has begun to diversify in this way. The ‘sociolinguistic typology’ movement recognises the need to integrate linguistic diversity with studies of socio-cultural process (Trudgill 2011; Di Garbo et al. 2021). Trudgill characterises this subfield as investigating ‘whether, and to what extent, the typological characteristics of the world’s languages are influenced by social structure and social organization – by the sociolinguistic characteristics of the communities in which they are spoken’ (Trudgill 2017: 124). A growing number of variationist sociolinguistic studies now investigate diverse linguistic settings, and especially the type of small societies in which most humans have lived for most of our history.

In this article I propose new connections between diverse variationist research and the evolutionary dynamics of language, with a particular focus on how social identity signalling may influence rates of change. I assemble an informal sample of variationist studies encompassing 63 languages from 26 families, and report on impressionistic patterns in this sample aligning types of linguistic structure with rates of change and social signalling. The main findings are that certain types of grammatical changes tend to occur quickly and may also be social signals (loss of redundant markers, paradigmatic mergers, changed form of markers), while others tend to change slowly and without social signalling (grammaticalisation of new markers, ordering variables). Phonological changes in general tend to be rapid, but lenition, deletion and assimilation appear to be more available as social signals, compared to phonemic mergers. Notably these observations depend not just on types of linguistic structures, such as segmental phonology, prosodic phonology, inflection etc, but additionally on *processes of change*, such as mergers, losses, gains. If these findings can be further substantiated, they may have significant implications for the evolutionary dynamics of language. However, for now this approach remains preliminary, and more extensive data is needed.

2. Evolutionary dynamics of language families

Many phylogenetic studies investigate the evolutionary dynamics of language by inferring which linguistic structures change faster or slower, and under what conditions (Greenhill et al. 2017). Some studies estimate rates-of-change for particular types of linguistic features (e.g. constituent order, stop voicing), but they may also investigate whether changes show a diversifying tendency, making related languages more different from one another than would be otherwise expected. Sometimes labelled ‘schismogenesis’ (after Bateson 1935), such changes are hypothesised to result from social groups using linguistic features to signal distinct group identities (e.g. Sorensen 1967; François 2011). When groups are in close contact, while also recognising each other as socially distinct, we might expect identity signalling to enhance the diversification of their languages or dialects. But just as different linguistic structures may in general change faster or slower, different structures may also have more or less utility for social identity signalling (Errington 1985; Cheshire 1987; Labov 1993; Eckert 2019).

Following early proposals by Nichols (1992; 1995), several studies have used phylogenetic methods to investigate rates of change of linguistic features, also often referred to as ‘(in)stability’ of features. Such studies vary in their specific methods and datasets, though many have used phonological and grammatical features from the World Atlas of Language Structures (WALS: Dryer & Haspelmath 2013). A meta-analysis shows a

promising degree of convergence between the results of these studies (Dediu & Cysouw 2013), for example finding slow change in some core inflectional features such as case marking and gender, as well as constituent order, but more rapid change in some phonological features such as consonant inventory size and consonant voicing, and in some pragmatically oriented grammatical features such as definite articles.¹ Individual features can also be grouped into types of linguistic structure defined by WALS, for example the Phonology category including features such as ‘Front Rounded Vowels’ and ‘Tone’, and the Word Order category including features such as ‘Order of Numeral and Noun’ and ‘Position of Polar Question Particles’. Differences of feature stability do not appear to generalise across these structural types (Greenhill et al. 2010; Greenhill et al. 2017), e.g. WALS Phonology features do not systematically differ from WALS Word Order features in their rates of change. However it remains possible that generalisations might hold over some alternative categorisation, and in the analysis below I will pay particular attention to types of change.

Some studies also measure stability of basic lexicon (changes of cognate sets), and compare this against structural features. An earlier study of Austronesian and Indo-European found no difference in rates of change between lexicon and structure (Greenhill et al. 2010), while a later study on Austronesian finds that most grammatical features change more rapidly than lexicon (Greenhill et al. 2017). This is somewhat surprising, since sociolinguists generally consider lexicon to be a prime domain for sociolinguistic differentiation (e.g. Durkin 2012), which might be expected to propel more rapid change.

A smaller set of studies investigate cross-linguistic evidence for schismogenetic change. Evans (2019) reviews selected examples from a wide range of languages, arguing that linguistic schismogenesis is more common than has been recognised. He presents some potential examples from the domains of morphology and syntax, but suggests that lexicon is likely the most frequent domain of schismogenesis, followed by phonetics. Turning to quantitative studies, phylogenetic reconstructions have been used to adduce evidence for increased rates of change when phylogenies split, often called ‘punctuational bursts’, which can be interpreted as a schismogenetic effect. A study of Austronesian, Bantu and Indo-European phylogenies found a positive correlation between the number of splits a language has been through and its number of lexical changes from the family root (Atkinson et al. 2008). A later study replicated this result for Austronesian, and additionally found a smaller burst effect for some WALS grammatical features, including word order and the presence of some grammatical categories (Greenhill et al. 2017). However this study does not systematically analyse the schismogenetic potential of different types of grammatical structure. Further results can soon be expected in this area, as phylogenetic methods have recently been extended to include punctuational bursts explicitly into computational models of language change (Douglas et al. 2025).

Using a different method, a correlational study of grammatical variables and dialect differentiation in 42 diverse languages finds that some types of variable are more likely distinguish dialects than others (Mansfield et al. 2023). The three types investigated are (1) form variables, in which distinct forms of a grammatical marker appear in the same linear position; (2) order variables, involving the same elements in different linear orders; and (3) omission variables, distinguished by the presence/absence of a grammatical marker.

¹ An alternative method tests stability of linguistic features using a metric that takes into account both vertical inheritance and spatial diffusion (Kauhanen et al. 2021). This further generalises the question of stability in linguistic transmission processes, and its results substantially converge with the earlier work, for example finding rapid change in phonological and pragmatic features (e.g. front rounded vowels; definite articles), and slow change in constituent order (e.g. object-verb, numeral-noun).

- (1) Kugu Nganhcara comitative (Smith & Johnson 2000: 393)
 a. *thuli-ra* b. *thuli-nta*
 woomera-COM woomera-COM
 ‘with a woomera’
- (2) Komnzo adjectival attribution (Döhler 2018: 89)
 a. *zagr karfo* b. *karfo zagr*
 distant village village distant
 ‘distant village’
- (3) Tundra Nenets comparison (Nikolaeva 2014: 174)
T'uku° pəni° taki° pəne-xəd° səwa(-rka)
 this coat that coat.ABL good(-COMP)
 ‘This coat is better than that one.’

They find that grammatical form variables have a high probability of differentiating neighbouring dialects, but order and omission variables both have a much lower probability. This supports earlier proposals made in a case study of Vanuatu languages (François 2011), suggesting that the form of grammatical markers may, like lexical forms, have schismogenetic utility, but linear ordering has little schismogenetic utility.

In summary, various types of cross-linguistic comparative research, some using phylogenetic methods or correlational tests, have yielded a substantial body of evidence on different rates of change among linguistic structures, and different capacities for sociolinguistic differentiation. Table 1 summarises the main findings by types of linguistic structure. In the following sections, I will discuss how these findings compare with results from a diverse sample of sociolinguistic variationist studies.

Table 1. Summary of expectations from comparative studies, and observations from diverse variationist studies.

LINGUISTIC STRUCTURE	RATE OF CHANGE AND SCHISMOGENETIC UTILITY
Grammatical categories	Core inflection: Slow change Pragmatic grammar: Rapid change
Ordering variables	Slow change, non-schismogenetic
Form of grammatical markers	<i>Not distinguished in studies of rates of change</i> Dialect differentiation study suggests schismogenetic effects
Phonology	Rapid change (but varies by phonological feature)
Lexicon	<u>Basic vocabulary</u> : Slow change, but highly schismogenetic

3. Diversity in variationist sociolinguistics

While the historical inferences of phylogenetic studies scope over millennia, an alternative approach is to study specific linguistic changes as they unfold over a few decades. This is the domain of variationist sociolinguistics, which takes advantage of the fact that linguistic changes spread via intra- and inter-speaker variation, as ‘different ways of saying the same thing’ (Weinreich et al. 1968). The main limitation of this field, from the point of view of evolutionary linguistics, has been its limited scope in terms of language diversity. With its gravitational centres in Britain and the United States, variationist sociolinguistics has been heavily focused on English and a few other European languages. For example in the decade to 2015, over half of the articles in *Language Variation and Change (LVC)* were on English alone, and 86% were on European languages (Stanford 2016). This limitation is not just about languages, but also about cultural contexts. Although variationism is deeply concerned with diversity in terms of gender, ethnicity and class, it has tended to focus on urban societies in a few wealthy nation states, while only occasionally venturing into the much wider range of sociolinguistic phenomena outside this context.

Fortunately, many variationists are well aware of this limitation, and the horizons of variationism are widening. There have been calls for a greater scope of diversity (Stanford 2016; Hildebrandt et al. 2017), and in 2015 the founding of a new journal, *Asia-Pacific Language Variation (APLV)*, promoting variationist research in an especially diverse region. These developments build on the earlier emergence of variationist research in non-European languages, such as Arabic (Royal 1985; Haeri 1994; Owens et al. 2013), Hebrew (Levon 2012; Gafter 2019), Mandarin (Zhang 2005), Cantonese (Zhang 2019), Japanese (Heffernan & Hiratuka 2018) and Indonesian (Gunarwan 1981; Abtahian et al. 2021). However the earlier studies were generally still on national languages with millions of speakers in urban settings. What has arisen more recently is a wave of research on ‘minority’ or ‘indigenous’ languages, that is to say, the majority of languages on earth, which also reflect the more diverse cultural settings humans inhabit outside of industrialised nation states. This is especially important if we hope to use variationist research as a window into broader patterns of language change.

The remainder of this article will give a taste of what is now available in diverse variationist research, and demonstrate the value of further expanding this field. I have assembled an informal sample of diverse variationist studies, selecting those that have most or all of the following features: (a) quantified use of linguistic variants; (b) social factors of variation; (c) a language that is not European, not Indo-European, and/or not a national language; (d) appear in a variationist journal. I constructed my sample by first going through content lists of *LVC* and *APLV* for the last two decades. I then used web searches and recommendations from colleagues to add some more studies, especially for under-represented regions. The resulting sample encompasses 63 languages from 26 families, including sign languages. Figure 1 illustrates language family and geographic coverage, while Table 2 provides the language names, families and study details. Numbers marked on the map correspond to listings in the table. Clearly, this is still a tiny sample of the approximately 8000 languages and 250 language families in the world. Furthermore, the coverage of regions and families is biased towards the Asia-Pacific region (thanks to *APLV*), and a strong representation of sign languages, while the regions of Africa, the Americas and northern Eurasia are under-represented. Thus the sample is small, and unbalanced. However given the lack of comparative research in sociolinguistics, this sample nonetheless reflects significant progress in the state of the art. In the following sections I will consider how the sociolinguistic variables reported in these studies shed light on hypotheses about the evolutionary dynamics of language.

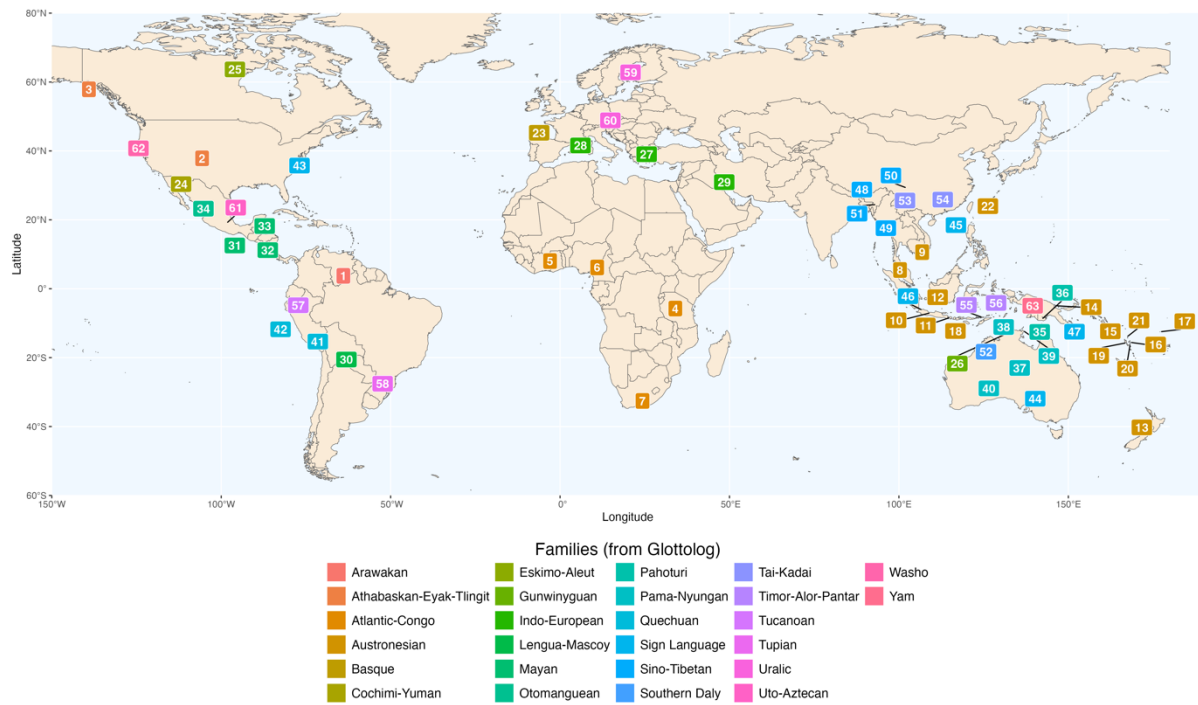


Figure 1. Language locations and family membership. Numbers on the map correspond to listings provided in Table 2.

Table 2. Details of languages and studies shown in Figure 1.

FAMILY	LANGUAGE	GLOTOCODE	STUDIES
Arawakan	1 Baniwa	bani1255	(Gonçalves 2018)
Athabaskan	2 Navajo	nava1243	(Palakurthy 2019)
	3 Tlingit	tlin1245	(Cardoso et al. 2022)
Atlantic-Congo	4 Chasu	asut1235	(Yohana 2009)
	5 Ewe	ewee1241	(Noglo 2009)
	6 Igbo	nucl1417	(Opara & Mbagwu 2021)
	7 Xhosa	xhos1239	(Bloom Ström 2018)
Austronesian	8 Acehnese	achi1257	(Travis & Ghina 2021)
	9 Eastern Cham	east2563	(Baclawski 2018)
	10 Javanese	java1254	(Zen & Starr 2021)
	11 Kata Kolok	bali1278	(Mudd et al. 2020)
	12 Madurese	nucl1460	(Misnadin 2021)
	13 Maori	maor1246	(Harlow et al. 2009)
	14 Matukar	matu1261	(Barth 2020; Davey & Barth 2024)
	15 Nkep	hogh1234	(Meyerhoff 2015)
	16 Raga	hano1246	(Duhamel 2022)
	17 Rotuman	rotu1241	(Fimone 2020)
	18 Sasak	sasa1249	(Khairunnisa 2021)
	19 Tamambo	malo1243	(Meyerhoff 2009)
	20 Vatlongos	sout2859	(Ridge 2022)
	21 Vera'a	vera1241	(Schnell & Barth 2018; Schnell & Barth 2020)
	22 Yami	yami1254	(Lai & Gooden 2024)

FAMILY		LANGUAGE	GLOTTOCODE	STUDIES
Basque	23	Basque	basq1248	(Gondra et al. 2024)
Cochimi-Yuman	24	Kumeyaay	kumi1248	(Field 2012)
Eskimo-Aleut	25	Inuit	inui1246	(Clarke 2009)
Gunwinyguan	26	Bininj Kunwok	gunw1252	(Marley 2021)
Indo-European	27	Albanian	alba1267	(Riverin-Coutlée et al. 2024)
	28	Faetar	fran1269	(Nagy 2009)
	29	Persian	west2369	(Ghafar Samar & Bhatia 2018)
Lengua-Mascoy	30	Sanapaná	sana1298	(Gysel 2022)
Mayan	31	K'ichee'	kich1262	(Romero 2009; Romero 2015)
	32	Kaqchikel	kaqc1270	(Leonard 2009)
	33	Tzeltal	tsel1254	(Rocío Cruz Gómez 2008)
Otomanguean	34	Chichimeca	chic1272	(Lastra 2009)
Pahoturi	35	Ende	ende1235	(Lindsey 2021; Strong et al. 2022)
	36	Idi	idii1243	(Schokkin 2021)
Pama-Nyungan	37	(Light) Warlpiri	warl1254	(O'Shannessy 2008)
	38	Djambarrpuyŋu	djam1256	(Wilkinson 1991)
	39	Gumatj	guma1253	(Amery 1985)
	40	Pitjantjatjara	pitj1243	(Wilmoth et al. 2021)
Quechuan	41	Chanka Quechua	ayac1238	(Povilonis & Guy 2022)
	42	Huaylas Quechua	huay1240	(Julca Guerrero 2010)
Sign Languages	43	American S.L.	amer1248	(Lucas et al. 2002)
	44	Auslan	aust1271	(Johnston et al. 2015)
	45	Hong Kong S.L.	hong1241	(Wei et al. 2018)
	46	Indonesian S.L.	indo1333	(Palfreyman 2020)
	47	Papua New Guinean S.L.	papu1255	(Reed 2020)
Sino-Tibetan	48	Angami	anga1288	(Suokhrie 2016)
	49	Black Lahu	lahu1253	(Yang et al. 2022)
	50	Ersu	ersu1241	(Chirkova et al. 2018)
	51	Manipuri	mani1292	(Satyanath & Laskar 2009)
Southern Daly	52	Murrinhpatha	murr1258	(Mansfield 2015b; Mansfield 2015a)
Tai-Kadai	53	Sui	suii1243	(Stanford 2009)
	54	Zhuang	yong1276	(Stanford & Pan 2013)
Timor-Alor-Pantar	55	Abui	abui1241	(Saad 2020)
	56	Fataluku	fata1247	(Grama et al. 2024)
Tucanoan	57	Máíhiŋi	orej1242	(Skilton 2017)
Tupian	58	Guaraní	para1311	(Shain & Tonhauser 2010)
Uralic	59	Finnish	finn1318	(Kuparinen et al. 2021)
	60	Hungarian	hung1274	(Csernicskó & Fenyvesi 2012)
Uto-Aztecan	61	Nahuatl	cent2132	(Vicente Ferre 2017)
Washo	62	Washo	wash1253	(Yu 2008)
Yam	63	Nmbo	namb1293	(Kashima 2020)

4. Implications of sociolinguistic variation for language evolution

Before diving into examples from diverse variationist studies, it is worth articulating some crucial parameters connecting linguistic variation, social structure and language change. Linguistic variables can in the first place be *correlated* with social categories (age, gender, class, etc), or they may not. A secondary question, for variants with social correlations, is whether people have meta-linguistic awareness of the variation, and/or meta-linguistic associations between linguistic variants and social categories. The nexus of meta-linguistic awareness and social categories is too complex to do justice to here (but see Campbell-Kibler 2016; Drager & Kirtley 2016), but when speakers have meta-linguistic awareness of a variable, they may also associate it with social categories, i.e. ‘social indexicality’ (Eckert 2019 *inter alia*). Two key questions for socio-linguistic theory are, what types of linguistic structures are more likely to have social correlations, and what types are more likely to be socio-indexical. Both questions can only be approached satisfactorily by taking into account a diverse range of language types and cultural contexts (Labov 2015).

Social correlations and social indexicality both have implications for evolutionary dynamics. When a variable correlates with speaker age, this may indicate rapid change in progress, perhaps unfolding within a few decades (Bailey et al. 1991), though it may also be a youth variant that people grow out of. When a variable furthermore has a social signalling function, this implies the possibility of schismogenetic change, where social groups favour distinct variants as a means of signalling their distinct group identities. Such variables can be expected to produce not just in more rapid language change, but also more language diversification, compared to other variables.

Regarding different types of linguistic structure, some discussion has been framed in terms of phonology, morphology and syntax (e.g. Lavandera 1978; Cheshire 1987), but these categories are likely to be too coarse. The slightly more granular nine-category classification of WALS variables has also been used, as described above, but this categorisation was not designed to reflect evolutionary dynamics. In the discussion below I will propose a different set of types, which crucially involve more attention to types of change, such as mergers, losses and gains. I developed these types heuristically as I searched for patterns in the sample.

5. Patterns emerging from diverse variationist studies

I now consider some potential patterns emerging from the sample of diverse variationist studies, and how these might align with the evolutionary dynamics described above. These should not be taken as strong claims, but rather as preliminary observations of a new research method. More robust generalisations will likely require a few hundred diverse variationist studies, rather than the few dozen consulted here. I begin with types of variation that appear to have the least connection to social categories, then move on to social correlations and finally social indexicality.

5.1. Grammaticalisation, redundant markers and mergers

The sample includes several instances where a grammatical category is variably expressed in morphology, which can be interpreted as cases of grammaticalisation-in-progress. One example is in Matukar (Austronesian), where event directionality is variably expressed with a suffix-like construction (Davey & Barth 2024). The suffix-like forms transparently reflect phrasal serial verb constructions (4a), which have undergone phonological integration and sharing of pronominal prefixes to produce a complex word construction (4b). This innovation does not show strong social conditioning and speaker awareness is not reported. The suffix-like variant is more frequent with some main verbs than others, and with some directional markers than others. This appears to be a change-in-progress largely independent of social dynamics, spreading through the whole speech community under the influence of linguistic

factors, and perhaps reiterating similar processes that have occurred in other Oceanic languages (Davey & Barth 2024: 180).

(4) Matukar

a. (serial verb construction)

...*ai luwadi-te diy-en di-mais-agokai*
 tree middle-LOC 3PL.S-live 3PL.S-**ascend**-I.R.IPFV.HAB
 ‘They live up in the middle of the tree.’ (Davey & Barth 2024: 162)

b. (suffix construction)

...*bom main di-tarutiti-pid-din-e*
 sago top 3PL.S-cut-**down**-3PL.P-D.SEQ
 ‘They cut down the sago...’

The Matukar example involves verb stems beginning to behave more like affixes. Other variables in the sample instead involve grammatical categories that can be redundantly marked. Such variables involve absence vs presence of a redundant grammatical marker, which have elsewhere been labelled ‘omission variables’ (Mansfield et al. 2023). One example is in Paraguayan Guaraní (Tupian: Shain & Tonhauser 2010), where suffixal marking of direct objects has been innovated (5a), extending the use of a marker *-pe/-me* that that was previously used only for recipients and locations (5b). This object marking remains variable, and is often redundant since the object role can be inferred from word order and/or context. The change has unfolded over a few centuries, and there is no mention of association with social categories. However it is likely to have been influenced by contact with Spanish, which has variable object marking favoured by animates. The Guaraní version is also favoured for animates, though it shows an additional preference for topical referents, which is not found in Spanish. Like with Matukar directional suffixes, the factors conditioning Guaraní object affixation appear to be purely linguistic, rather than social.

(5) Guaraní

a. (innovative object marking)

Kuehe a-hecha peteĩ kuñá-me
 yesterday B1.SG-see one woman-**PE**
 ‘Yesterday I saw a woman.’ (Shain & Tonhauser 2010)

b. (traditional locative usage)

Che-váje-pe o-ĩ va’ekue peteĩ karai
 B1.SG-town-**PE** A3-be back.then one gentleman
 ‘In my town, there used to be a gentleman.’

There are further examples of grammatical affixation developing seemingly without strong social conditioning or speaker awareness. Another omission variable is negative concord in K’ichee’ Mayan, again potentially influenced by contact with Spanish (Romero 2015). A more complex example comes from Idi (Pahoturi), where a verb-final suffix *-n* can be used for both 1SG and 3SG subjects, and depends on conjugation class (Schokkin 2021). Although speakers are reported to be largely unaware of variation in this suffix, younger speakers use it more than older speakers, which could eventually point to its generalisation as a 1SG/3SG marker.

In other instances, redundant grammatical markers are being lost from a language within a couple of generations. For example in Oxchuc Tzeltal, progressive aspect is redundantly marked by two particles, of which the second can be omitted (Rocío Cruz Gómez 2008: 63). There is no mention of meta-linguistic awareness of this variation, but

younger speakers omit the particle more frequently than older speakers, as much as 88% omission in casual speech (Rocío Cruz Gómez 2008: 118), which foreshadows a relatively rapid loss of the marker. In another Oxchuc Tzeltal omission variable, noun classifiers can be omitted with non-human referents, when the lexical noun is present. This variable is reported to induce meta-linguistic awareness, with classifier omission considered to be incorrect speech (Rocío Cruz Gómez 2008: 66). While grammatical omission variables do not usually induce meta-linguistic awareness, perhaps noun classifiers are more salient to speakers since they have relatively concrete meanings, like lexical content words (Berlin & Romney 1964).

Grammatical markers can also be rapidly lost when mergers occur in inflectional paradigms. We find here another example involving noun classifiers, this time in Chichimec (Otomanguan), where four noun classes are marked on possessive pronouns: Food, Clothes, Animals, Other. Younger speakers largely collapse the Other class into the erstwhile Food class, creating a three-class system: Clothes, Animals, Other (Lastra 2009: 168). The merger is almost complete,² and has occurred rapidly over a few generations, though, there is no mention of meta-linguistic awareness. Another paradigmatic merger is in Vatlongos (Austronesian), where subject prefixes traditionally distinguish singular, dual, paucal and plural number categories (Ridge 2022: 83). Table 3 shows some relevant forms for the immediate future tense. In the community of Mele Maat, which is closer to the urban centre than other Vatlongos communities and has more intensive contact with the lingua franca Bislama (Ridge 2022: 74), the paucal (PC) category appears to be merging in to the plural category. Again the change appears to occur rapidly and with no report of speaker awareness.

Table 3. Vatlongos immediate future subject prefixes (Ridge 2022: 83).

	Traditional			Mele Maat		
	1	2	3	1	2	3
SG	<i>na-</i>	<i>o-</i>	<i>va-</i>	<i>na-</i>	<i>a-</i>	<i>va-</i>
DU	<i>malo-</i>	<i>mulo-</i>	<i>lalo-</i>	<i>malo-</i>	<i>mulo-</i>	<i>lalo-</i>
PC	<i>mato-</i>	<i>muto-</i>	<i>lato-</i>			
PL	<i>ma-</i>	<i>mu-</i>	<i>la-</i>	<i>ma-</i>	<i>mu-</i>	<i>la-</i>

In summary, variables involving grammaticalisation, omission of grammatical markers, and mergers of inflectional paradigms are not reported to induce meta-linguistic awareness, and therefore can be expected to play less of a role in schismogenetic change. The development of new grammatical markings is slow enough that differences between age cohorts tend to be slight; but the loss of redundant markers, or merger of paradigmatic categories, can both happen within a couple of generations.

5.2. Ordering variables

There is a notable scarcity of ordering variables in the sample. This would appear to prevent us from saying anything about this type, but the absence could also be taken as meaningful: the lack of such variables, among studies that investigate sociolinguistic variation, may imply that ordering variables rarely have social correlations. On the other hand, it is also possible that variationist linguists tend to miss ordering variables, due to a preconception that they are not socially correlated (but see e.g. Schwenter & Torres Cacoullos 2010; Haddican et al. 2020).

There is only one ordering variable in the sample, which involves variable suffix ordering in Murrinhpatha (Southern Daly) (Mansfield 2015a). This appears to be changing within the period of a few generations, suggesting a rapid change. However it is unclear

² Except for a few lexemes denoting saints and sacred items, which continue to be classed by younger speakers in the erstwhile Other class, which would make this a micro-class for Sacred referents.

whether this is particular to affix order as opposed to word order, which is more frequently observed in family-level language change. Overall, the sparse attestation of socially correlated ordering variables remains compatible with the expectation from evolutionary dynamic studies that word order usually changes slowly, and without social influence.

5.3. *Phonemic mergers*

Another type of change that occurs frequently in the sample is phonological mergers, where some speakers tend to collapse a phonemic contrast maintained by others. Features neutralised include aspiration (Zhuang, Tai-Kadai; Stanford & Pan 2013; Madurese, Austronesian; Misnadin 2021), retroflexion (Javanese, Austronesian; Zen & Starr 2021), velar labialisation (Igbo, Atlantic-Congo; Opara & Mbagwu 2021; Máihiki, Tukanoan; Skilton 2017) and rounded front vowels (Chichmec, Otomanguean; Lastra 2009). Most of these involve rapid changes across a few generations, and in some instances the merger is associated with a specific regional dialect. Language contact may again play a role, for example Madurese speakers in the western region have more contact with Javanese, and are losing the Madurese retroflex coronal distinction, which is absent in Javanese (Zen & Starr 2021). But while phonemic mergers can happen quickly, none of the studies report meta-linguistic awareness. For aspiration mergers in Zhuang (Stanford & Pan 2013), which is a stable dialectal variable, it is specifically noted that speakers do not find the variation noticeable. Overall, it appears that phonemic mergers can be rapid changes, which may result in differences between closely related varieties, but there is no evidence of mergers as schismogenetic change.

5.4. *Form of grammatical markers*

Several variables involve distinct phonological forms of a grammatical marker (e.g. an affix, clitic or particle), which involve a morphologically specific sound-change (Mansfield et al. 2023). These variables thus have a phonological aspect, but also a grammatical aspect. They don't necessarily affect grammatical structure or grammatical categories, but they have potential to do so, for example if they later facilitate category mergers. The grammatical form variables in the sample tend to have social category correlations, and in some instances provoke meta-linguistic awareness.

One example is in Vatlongos (Austronesian), where verbal 2SG subject prefixes have alternate forms differentiated by an *a* > *o* innovation, e.g. *ote-* ~ *ate-* 2SG.PRIOR and *o-* ~ *a-* 2SG.NFUT (Ridge 2022). There is evidence that this is a rapid change-in-progress led by women, and speakers have meta-linguistic awareness of the variation, which they associate with the more urbanised community of Mele Maat, in contrast to the more traditional communities (Ridge 2022: 86). Similarly, in Oxchuc Tzeltal, definiteness is marked by a combination of proclitic and enclitic, but these have alternative forms (to simplify somewhat) *te*=[NP]=*e* ~ *i*=[NP]=*i* (Rocío Cruz Gómez 2008: 62). Speakers of all ages show style-shifting between the former variant in monitored speech and the latter in unmonitored speech, suggesting that these may be metalinguistically evaluated as correct and incorrect variants respectively (Rocío Cruz Gómez 2008: 144). However not all grammatical form variables provoke meta-linguistic awareness. In Máihiki (Tukanoan) there is variation between past tense suffixes, e.g. *-ao* ~ *-go* '3SG.F.PST', which reflect parental dialects of the speakers, though in this case the variable is not the focus of meta-linguistic awareness (Skilton 2017: 102–108). Overall, grammatical form variables appear liable to rapid change, and are sometimes metalinguistically recognised as differentiating social groups.

5.5. Phonological reduction (lenition, deletion and assimilation)

The most frequent type of variable in the sample is phonological reduction. In most cases (though not all), younger people use a phonologically reduced variant more frequently than older speakers. In several cases this also induces meta-linguistic awareness, including negative evaluation from older speakers.

In Pitjantjatjara (Pama-Nyungan), there is a variable vowel deletion process targeting extrametrical vowels at word boundaries, which appears to have been active in the community for some time (Goddard 1985; Wilmoth 2022: 148). This is illustrated in (6a), where the word-final /i/ is external to the foot formed by the preceding two syllables. However younger speakers show signs of extending deletion to syllables that could fall within feet (note that codas are not moraic), as in (6b). Furthermore, a quantitative study of deletion in the specific context of verb-final tense suffixes shows that younger speakers here systematically extend deletion beyond the metrical constraint observed in older speakers (6c). Vowel deletion is sociolinguistically salient, with older speakers criticising younger speakers for their ‘short’ (*mutumutu*) pronunciation (Wilmoth et al. 2021). This is despite the fact that older speakers also produce deleted forms, though perhaps reflecting sensitivity to the more constrained contexts applied by older speakers, compared to younger speakers.

(6) Pitjantjatjara

- a. *A-na-ny(i)* *tjuwa-kutu*
go-AUG-PRS store-ALL
‘Go to the store.’
- b. *Tjuḷp(u)* *panya*
bird ANAPH
‘That bird.’ (Wilmoth 2022: 118–119)
- c. *Kunkun-ari-ŋ(u)*
sleep-INCH-PST
‘Was beginning to sleep.’ (Wilmoth et al. 2021)

Several other instances involve word-initial reduction in Australian Aboriginal languages, including /p, k/ lenition in Murrinhpatha (Southern Daly; Mansfield 2015b), /ŋ/ deletion in Bininj Kun-wok (Gunwinjguan; Marley 2021) and /ŋ/ deletion in Gumatj (Pama-Nyungan; Amery 1985: 45). In some instances there is evidence that initial reduction is part of a sound-change in progress (Marley 2021), others appear to involve age-graded style shifting (Langlois 2006), but in many studies there is insufficient evidence to decide between change-in-progress versus stable age-grading. This is the case for Murrinhpatha initial lenition, where older speakers prefer stop realisations of initial /p, k/, while younger speakers often produce fricative realisations or other lenited variants (7). Older Murrinhpatha speakers identify stop variants as the correct forms, which some describe as *bangamlele* or ‘biting’ the words, but nonetheless, many younger speakers do not show any sign of style-shifting. This variable may also be related to contact with Western Daly languages, where initial obstruents vary freely between stop and fricative articulations (Mansfield & Green 2021).

(7) Murrinhpatha (young speakers)

- a. *purtek* [puṭek ~ ɸuṭek]
ground
- b. *kardu* [kaḍu ~ xaḍu]
person (Mansfield 2015b: 206–209)

Beyond Australia, in Ende (Pahoturi) retroflex obstruents are frequently affricated in everyday speech as [tʂ, dʂ], but prestigious individuals with status as public orators prefer

purely plosive variants [t, d] (Strong et al. 2022). In Nmbo (Yam) younger speakers favour initial /h/ deletion, which appears to be a rapid change-in-progress (Kashima 2020). In Australian and New Zealand Sign language, signs that are canonically produced at the forehead have lowered variants, which can be interpreted as a type of phonological reduction (reduced effort), and these variants are favoured by younger signers (Schembri et al. 2009). Other changes-in-progress are at or near completion, with young people almost completely shifted to reduced variants. For example in Oxchuc Tzeltal (Mayan), unstressed vowel deletion is highly frequent in younger speakers (Rocío Cruz Gómez 2008: 126), while in Chichimec (Otomanguan), initial /g/ reaches categorical deletion in young speakers (8) (Lastra 2009).³ In Inuit (Eskimo-Aleut), initial short vowel deletion approaches categorical deletion in young speakers (9) and is subject to style-shifting, likely forming part of the recognised youth speech style that is bemoaned by older speakers (Clarke 2009: 112, 120).

(8) Chichimec

- a. (g)a-kén
1.FUT-pull
- b. (g)a-rác?
3.FUT-rub (Lastra 2009: 162)

(9) Inuit

- a. (a)kuhp
coat
- b. (i)skeu
woman (Clarke 2009: 122)

Not all reduction variants are straightforwardly favoured by younger speakers as opposed to older speakers. In Máihiki (Tukanoan), intervocalic /h/ deletion has regional dialect associations, and some speakers consider it perfectly acceptable while others consider it to be incorrect (Skilton 2017). In Fataluku (Timor-Alor-Pantar), the fricative /z/ is lenited to a glide in word-medial position, with a traditional rural dialect association (Grama et al. 2024). Raga (Austronesian) has initial /g/ deletion, favoured by both younger men and older men, potentially linked by their kava-drinking socialisation (Duhamel 2022). In Rotuman (Austronesian), older speakers retain /ʔ/ while middle-aged speakers it, which may reiterate glottal-stop losses found in Oceanic languages (Fimone 2020). However in this instance, younger speakers, who are deeply concerned about Rotuman language and identity, retain /ʔ/ as a sociolinguistically salient marker of being Rotuman in the face of cultural homogenisation.

Alongside phonological reduction, the sample includes a few instances of phonological assimilation, where one segment is variably influenced by neighbouring segments. We might expect these to pattern with reduction variables, due to the general link between reduction and segmental context (Harris 1984; Katz 2016), and indeed the evidence suggests that assimilation variables are also amenable to socioindexicality. For example Eastern Cham has variably labialised nasal codas after rounded vowels, e.g. [uŋ ~ uŋm] (Baclawski 2018). The labialised variant likely derives from contact with Vietnamese, which has similar allomorphy. Earlier in the twentieth century the labialised variant was associated with women's speech, though it now appears to be spreading as a change among all speakers.

Overall, the multiple instances of phonological reduction (and assimilation) variables in diverse languages suggest that this is a very frequent form of sociolinguistic variation,

³ Chichimec /g/ only occurs word-initially in a few verbal prefixes, though it occurs medially in a variety of words.

often sociolinguistically salient, and often associated with youth speech. Since reduced articulation is present to some extent in all human language, it provides an ever-present means for younger people to annoy their elders. In some instances, youth reduction variants may become sound changes, and while this would apparently be favoured in situations where the reduction is unconscious, there is an intriguing possibility that salient reduction variables may also become sound changes, if older people are not able to impose a stylistic switch to phonologically conservative variants. At the same time, even if there is a tendency for reduction variants to be preferred by youth, this is not an inviolable rule, and other sociolinguistic dynamics may unfold. This may be due to language contact, or as we saw in Rotuman, ethno-linguistic identity becoming attached to a conservative variant.

5.6. Lexicon

Although lexicon is expected to be one of the main dimensions of schismogenetic change, there is little to report on lexicon in the sample. In fact lexicon generally tends to be excluded from variationist studies for methodological reasons (Durkin 2012). The sample includes one dedicated study of lexical variation in Kata Kolok, a Balinese sign language (Mudd et al. 2020), though this focuses on the degree of lexical variation within deaf vs hearing groups, and is not easily relatable to questions of evolutionary dynamics. Studies on dialectal variation in Sui and Zhuang (both Tai-Kadai) also describe some lexical variation, in both cases finding strong social indexicality as expected. In Sui, women move to their husband's village, where they are expected to maintain sociolinguistic markers of their own clan, especially lexicon (Stanford 2009). For example, women from the Pan group maintain use of a distinctive 1SG pronoun *ju*, after moving to a Zhang village where the local 1SG pronoun is *ej*. Shifting to the local pronoun would be considered an act of disloyalty to their own ethno-geographic heritage. By contrast, in Zhuang, where wives also move to their husband's village, women from the northern dialect group who marry into the southern dialect group are expected to shift to the lexicon of their husband's group (Stanford & Pan 2013). Whether the rules are for lexical differentiation, or lexical shift, these dialect studies both demonstrate the importance of the lexicon for social signalling.

6. Discussion and conclusion

As we have seen, the increasingly diverse variationist literature reveals potential trends in the evolutionary dynamics of language. This offers an alternative approach to the existing tradition of phylogenetic studies, and in general the findings converge with that tradition on several points. Table 4 repeats the main findings gleaned from phylogenetic and correlational studies above, now presented alongside patterns observed in the variationist sample.

Table 4. Summary of findings from phylogenetic and correlational studies, and from diverse variationist studies.

VARIABLE TYPE	PHYLO/CORRELATIONAL FINDINGS	VARIATIONIST FINDINGS
Grammatical categories	<u>Core inflection</u> : Slow change <u>Pragmatic grammar</u> : Rapid change	<u>Grammaticalisation</u> : Slow <u>Redundant marker loss</u> : Rapid <u>Paradigmatic merger</u> : Rapid
Ordering variables	Slow change, non-schismogenetic	<i>Limited evidence</i> , but compatible with slow change, non-schismogenetic
Form of grammatical markers	<i>Not distinguished in studies of rates of change</i> Dialect differentiation study suggests schismogenetic effects	Potential for rapid change and schismogenesis
Phonology	Rapid change (but varies by phonological feature)	<u>Phonemic mergers</u> : Rapid change, non-schismogenetic <u>Reduction and assimilation</u> : Rapid change, schismogenetic

Lexicon	<u>Basic vocabulary</u> : Slow change, but highly schismogenetic	<i>Limited evidence</i> , but studies report schismogenetic effects
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Regarding grammatical categories, the earlier studies do not reach any strong overall generalisations, but appear to show differences depending on type of category. In the variationist studies just reviewed, potential differences were noted according to type of *process*, i.e. whether markers are being gained, lost, or merged. The evidence suggest that new markers emerge slowly, but old markers can be lost quickly if they are either informationally redundant, or susceptible to a paradigmatic merger. Regarding ordering variables, both approaches broadly agree that these change slowly and without schismogenetic effects. The form of grammatical markers has not been investigated in phylogenetic studies, but the dialect differentiation study finds potential schismogenetic effects, which are echoed in some variationist studies.

In phonology, the variationist studies again suggest differences according to the type of process, which are not distinguished in the phylogenetic studies. Both types of study suggest that at least some types of phonology can change rapidly, and variationist studies additionally suggest that phonological reduction and assimilation tend to provoke metalinguistic awareness, but phonemic mergers do not tend to provoke awareness. Language contact is potentially implicated in a high proportion of instances, for both grammatical and phonological variables. This suggests that in small societies, where multilingualism is the norm (Lüpke 2016), contact may play an especially prominent role in language change.

As for lexicon, variationist research does not add much to the existing picture, but what evidence we do find remains compatible with the idea that lexicon has a high schismogenetic utility. This may seem paradoxical alongside the phylogenetic finding that lexicon changes more slowly than grammatical features (Greenhill et al. 2017), since schismogenetic processes in general should drive more rapid change. One potential resolution could be that schismogenetic lexical change does not tend to target the same basic lexicon that is investigated in phylogenetic studies. Another possibility is that the number of changes involved in a rapid, punctuational burst of lexical change is simply less than the number of grammatical featural changes built up slowly over a process of gradual change.

Clearly, much more research is required on the evolutionary dynamics of language. However between the variationist and phylogenetic approaches, we can hope to gradually triangulate more findings. A similar type of triangulation has recently become popular between typological findings and psycholinguistic experiments (Sinnemäki 2014), for example showing that typologically preferred structures are also preferred by experimental participants in the domains of noun-phrase order (Culbertson et al. 2012; Culbertson et al. 2020) and affix order (Saldana et al. 2021; Mansfield et al. 2022). For the moment, triangulation between variationism and phylogenetics remains much more preliminary, and the variationist case studies are still too sparse for a systematic meta-analysis. But the tentative patterns observed above should motivate further growth in this area. Ideally, the surge of variationist study in the Asia-Pacific region will be matched by similar growth in other regions, eventually producing a more detailed picture of linguistic dynamics in diverse speech communities.

References

- Abtahian, Maya Ravindranath & Cohn, Abigail C. & Djenar, Dwi Noverini & Vogel, Rachel C. 2021. Jakarta Indonesian first-person singular pronouns: Form, function and variation. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 7(2). 185–214. (doi:10.1075/aplv.20012.rav)
- Amery, Robert. 1985. *A new diglossia: Contemporary speech varieties at Yirrkala in north east Arnhem Land*. Canberra: Australian National University. (Masters thesis.)

- Atkinson, Quentin D. & Meade, Andrew & Venditti, Chris & Greenhill, Simon J. & Pagel, Mark. 2008. Languages evolve in punctuational bursts. *Science*. American Association for the Advancement of Science. (doi:10.1126/science.1149683) (<https://www.science.org/doi/abs/10.1126/science.1149683>) (Accessed February 3, 2022.)
- Baclawski, Kenneth. 2018. Diglossia and change from below in Eastern Cham. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 4(1). 73–102. (doi:10.1075/aplv.17003.bac)
- Bailey, Guy & Wikle, Tom & Tillery, Jan & Sand, Lori. 1991. The apparent time construct. *Language Variation and Change* 3(3). 241–264. (doi:10.1017/S0954394500000569)
- Barth, Danielle. 2020. Variation in Matukar Panau kinship terminology. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 5(2). 138–170. (doi:10.1075/aplv.00004.bar)
- Bateson, Gregory. 1935. Culture contact and schismogenesis. *Man*. [Wiley, Royal Anthropological Institute of Great Britain and Ireland] 35. 178–183. (doi:10.2307/2789408)
- Berlin, Brent & Romney, A. Kimball. 1964. Descriptive semantics of Tzeltal numeral classifiers. *American Anthropologist* 66(3). 79–98. (doi:10.1525/aa.1964.66.3.02a00840)
- Bloom Ström, Eva-Marie. 2018. Linguistic and Sociolinguistic Aspects of Variation in the Eastern Cape: Complexities of Xhosa Language Use. *Studia Orientalia Electronica* 6. 90–120. (doi:10.23993/store.75138)
- Bromham, Lindell. 2024. Population size and language change: An evolutionary perspective. *Annual Reviews*. (doi:10.1146/annurev-linguistics-031422-123959) (<https://www.annualreviews.org/content/journals/10.1146/annurev-linguistics-031422-123959>) (Accessed January 7, 2025.)
- Campbell-Kibler, Kathryn. 2016. Towards a cognitively realistic model of meaningful sociolinguistic variation. In Babel, Anna M. (ed.), *Awareness and Control in Sociolinguistic Research*, 123–151. Cambridge: Cambridge University Press. (doi:10.1017/CBO9781139680448.008)
- Cardoso, Amanda & Crippen, James & Mellesmoen, Gloria. 2022. Cross-dialectal synchronic variation of a diachronic conditioned merger in Tlingit. *Linguistics Vanguard*. De Gruyter Mouton 8(s5). 519–530. (doi:10.1515/lingvan-2021-0048)
- Cheshire, Jenny. 1987. Syntactic variation, the linguistic variable, and sociolinguistic theory. *Linguistics* 25(2). 257–282. (doi:10.1515/ling.1987.25.2.257)
- Chirkova, Katia & Stanford, James N. & Wang, Dehe. 2018. A long way from New York City: Socially stratified contact-induced phonological convergence in Ganluo Ersu (Sichuan, China). *Language Variation and Change* 30(1). 109–145. (doi:10.1017/S095439451700028X)
- Clarke, Sandra. 2009. Sociolinguistic stratification and new dialect formation in a Canadian aboriginal community. In Stanford, James N. & Preston, Dennis Richard (eds.), *Variation in Indigenous Minority Languages*. Amsterdam: John Benjamins.
- Csernicskó, István & Fenyvesi, Anna. 2012. Sociolinguistic and Contact-induced Variation in Hungarian Language Use in Subcarpathia, Ukraine. *AHEA: E-journal of the American Hungarian Educators Association* 5. 1–30.
- Culbertson, Jennifer & Franck, Julie & Braquet, Guillaume & Barrera Navarro, Magda & Arnon, Inbal. 2020. A learning bias for word order harmony: Evidence from speakers of non-harmonic languages. *Cognition* 204. 104392. (doi:10.1016/j.cognition.2020.104392)
- Culbertson, Jennifer & Smolensky, Paul & Legendre, Géraldine. 2012. Learning biases predict a word order universal. *Cognition* 122(3). 306–329. (doi:10.1016/j.cognition.2011.10.017)
- Davey, Kira & Barth, Danielle. 2024. Directional constructions in Matukar Panau: A Bayesian approach to assessing variation. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 9(2). 156–194. (doi:10.1075/aplv.23005.dav)
- Dediu, Dan & Cysouw, Michael. 2013. Some structural aspects of language are more stable than others: A comparison of seven methods. *PLOS ONE* 8(1). e55009. (doi:10.1371/journal.pone.0055009)
- Di Garbo, Francesca & Kashima, Eri & Napoleão de Souza, Ricardo & Sinnemäki, Kaius. 2021. Concepts and methods for integrating language typology and sociolinguistics. In Ballarè, Silvia & Inglese, Guglielmo (eds.), *Tipologia e Sociolinguistica* (Nuova Serie), 143–176. Milano: Officinaventuno. (doi:10.17469/O2105SLI000005)
- Döhler, Christian. 2018. *A grammar of Komnzo*. Berlin: Language Science Press. (doi:10.5281/zenodo.1477799) (<http://langsci-press.org/catalog/book/212>) (Accessed April 19, 2019.)
- Douglas, Jordan & Bouckaert, Remco & Harris, Simon C. & Carter Jr., Charles W. & Wills, Peter R. 2025. Evolution is coupled with branching across many granularities of life. *Proceedings of the Royal Society B: Biological Sciences*. Royal Society 292(2047). 20250182. (doi:10.1098/rspb.2025.0182)
- Drager, Katie & Kirtley, M. Joelle. 2016. Awareness, salience, and stereotypes in exemplar-based models of speech production and perception. In Babel, Anna M. (ed.), *Awareness and Control in Sociolinguistic Research*, 1–24. Cambridge: Cambridge University Press. (doi:10.1017/CBO9781139680448.003)

- Dryer, Matthew S. & Haspelmath, Martin (eds.). 2013. *WALS Online*. Leipzig: Max Planck Institute for Evolutionary Anthropology. (<https://wals.info/>) (Accessed November 4, 2022.)
- Duhamel, Marie-France. 2022. The role of older men in a phonological change: (y) in Raga, Vanuatu. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 8(2). 206–239. (doi:10.1075/aplv.21002.duh)
- Durkin, Philip. 2012. Variation in the lexicon: the ‘Cinderella’ of sociolinguistics?: Why does variation in word forms and word meanings present such challenges for empirical research? *English Today* 28(4). 3–9. (doi:10.1017/S0266078412000375)
- Eckert, Penelope. 2019. The limits of meaning: Social indexicality, variation, and the cline of interiority. *Language* 95(4). 751–776. (doi:10.1353/lan.2019.0072)
- Enfield, Nick J. 2014. *Natural causes of language: Frames, biases and cultural transmission*. Berlin: Language Science Press.
- Errington, J. Joseph. 1985. On the nature of the sociolinguistic sign: Describing the Javanese speech levels. In Mertz, Elizabeth & Parmentier, Richard J. (eds.), *Semiotic mediation: Sociocultural and psychological perspectives*, 287–310. New York: Academic Press.
- Evans, Nicholas. 2018. The dynamics of language diversity. In Mesthrie, Rajend & Bradley, David (eds.), *The dynamics of language: Plenary and focus lectures from the 20th International Congress of Linguists*, 12–41. Cape Town: University of Cape Town.
- Evans, Nicholas. 2019. Linguistic divergence under contact. In Cennamo, Michela & Fabrizio, Claudio (eds.), *Historical Linguistics 2015: Selected papers from the 22nd International Conference on Historical Linguistics*, 564–591. Amsterdam: John Benjamins.
- Field, Margaret. 2012. Kumeyaay Language Variation, Group Identity, and The Land. *International Journal of American Linguistics*. The University of Chicago Press 78(4). 557–573. (doi:10.1086/667451)
- Fimone, Wilfred. 2020. Onset glottal stop deletion in Suva Rotuman. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 6(2). 196–221. (doi:10.1075/aplv.19016.fim)
- François, Alexandre. 2011. Social ecology and language history in the northern Vanuatu linkage: A tale of divergence and convergence. *Journal of Historical Linguistics*. John Benjamins 1(2). 175–246. (doi:10.1075/jhl.1.2.03fra)
- Gafter, Roey J. 2019. Modern Hebrew Sociophonetics. *Brill’s Journal of Afroasiatic Languages and Linguistics*. Brill 11(1). 226–242. (doi:10.1163/18776930-01101014)
- Ghafar Samar, Reza & Bhatia, Tej. 2018. The future of ‘future’: A Persian perspective on grammaticalization of future marking. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 3(2). 130–159. (doi:10.1075/aplv.16011.gha)
- Goddard, Cliff. 1985. *A grammar of Yankunytjatjara*. Alice Springs: Institute for Aboriginal Development.
- Goldstein, David M. 2022. There’s no escaping phylogenetics. In Grestenberger, Laura & Reiss, Charles & Fellner, Hannes A. & Pantillon, Gabriel Z. (eds.), *Ha! Linguistic studies in honor of Mark R. Hale*, 71–91. Wiesbaden: Reichert.
- Gonçalves, Artur Garcia. 2018. *Para uma dialetologia baniwa-koripako do rio Içana*. Brasília: Universidade de Brasília. (Masters thesis.)
- Gondra, Ager & Rodríguez-Ordóñez, Itxaso & Franco-Landa, Eukene. 2024. Variation in the production of Basque ergativity: Change or stable variation? *Language Variation and Change* 36(1). 95–120. (doi:10.1017/S0954394524000048)
- Grama, James & Heston, Tyler M. & Ross, Melody Ann. 2024. Variation in the Fataluku voiced coronal (j). *Asia-Pacific Language Variation*. John Benjamins Publishing Company 9(2). 125–155. (doi:10.1075/aplv.00017.gra)
- Gray, Russell D. & Greenhill, Simon J. & Atkinson, Quentin D. 2013. Phylogenetic models of language change: Three new questions. *Cultural evolution: Society, technology, language and religion*, 285–300. Cambridge, MA: MIT Press.
- Greenhill, Simon J. 2015. Demographic correlates of language diversity. In Bower, Claire & Evans, Bethwyn (eds.), *The Routledge Handbook of Historical Linguistics*, 557–578. Abingdon: Routledge.
- Greenhill, Simon J. & Atkinson, Quentin D. & Meade, R. & Gray, Russell D. 2010. The shape and tempo of language evolution. *Proceedings of the Royal Society B: Biological Sciences* 277(1693). 2443–2450. (doi:10.1098/rspb.2010.0051)
- Greenhill, Simon J. & Wu, Chieh-Hsi & Hua, Xia & Dunn, Michael & Levinson, Stephen C. & Gray, Russell D. 2017. Evolutionary dynamics of language systems. *Proceedings of the National Academy of Sciences* 114(42). E8822–E8829. (doi:10.1073/pnas.1700388114)
- Gunarwan, Asim. 1981. *A sociolinguistic study of grammatical variation in Indonesian*. Washington, D.C: Georgetown University. (PhD thesis.)
- Gysel, Jens E. L. Van. 2022. The influence of language shift on Sanapaná vowels: An exemplar-based perspective. *Language Variation and Change* 34(1). 107–135. (doi:10.1017/S0954394522000023)

- Haddican, Bill & Johnson, Daniel Ezra & Wallenberg, Joel & Holmberg, Anders. 2020. Variation and change in the particle verb alternation across English dialects. In Beaman, Karen V. & Buchstaller, Isabelle & Fox, Sue & Walker, James A. (eds.), *Advancing socio-grammatical variation and change*, 15–31. London: Routledge.
- Haeri, Niloofar. 1994. A linguistic innovation of women in Cairo. *Language Variation and Change* 6(1). 87–112. (doi:10.1017/S0954394500001599)
- Harlow, Ray & Keegan, Peter & King, Jeanette & Maclagan, Margaret & Watson, Catherine. 2009. The changing sound of the Maori language. In Stanford, James N. & Preston, Dennis Richard (eds.), *Variation in Indigenous Minority Languages*, 129–152. Amsterdam: John Benjamins.
- Harris, James. 1984. La espirantiación en castellano y la representación fonológica autosegmental. *Estudios gramaticales* 1. 149–167.
- Heffernan, Kevin & Hiratuka, Yusuke. 2018. Morphological relative frequency impedes the use of stylistic variants: Evidence from a corpus of vernacular Japanese. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 3(2). 200–231. (doi:10.1075/aplv.16009.hef)
- Hildebrandt, Kristine A. & Jany, Carmen & Silva, Wilson. 2017. Introduction: Documenting variation in endangered languages. In Hildebrandt, Kristine A. & Jany, Carmen & Silva, Wilson (eds.), *Documenting Variation in Endangered Languages* (Language Documentation & Conservation Special Publication 13), 1–5. Honolulu: University of Hawai'i Press.
- Johnston, Trevor & Cresdee, Donovan & Schembri, Adam & Woll, Bencie. 2015. FINISH variation and grammaticalization in a signed language: How far down this well-trodden pathway is Auslan (Australian Sign Language)? *Language Variation and Change* 27(1). 117–155. (doi:10.1017/S0954394514000209)
- Julca Guerrero, Felix Claudio. 2010. *Variación dialectal del Quechua en el Callejón de Huaylas desde las perspectivas de la lingüística y de los hablantes*. Austin, TX: University of Texas at Austin. (PhD Thesis.) (<http://hdl.handle.net/2152/ETD-UT-2010-05-1020>) (Accessed January 22, 2025.)
- Kashima, Eri. 2020. Word-initial [h]-drop variation in Nmbo: Change-in-progress in an egalitarian multilingual speech community of Papua New Guinea. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 6(2). 250–277. (doi:10.1075/aplv.20002.kas)
- Katz, Jonah. 2016. Lenition, perception and neutralisation. *Phonology* 33(1). 43–85.
- Kauhanen, Henri & Gopal, Deepthi & Galla, Tobias & Bermúdez-Otero, Ricardo. 2021. Geospatial distributions reflect temperatures of linguistic features. *Science Advances*. American Association for the Advancement of Science. (doi:10.1126/sciadv.abe6540)
- Khairunnisa. 2021. Variation in Ampenan Sasak pronominal forms. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 7(2). 120–141. (doi:10.1075/aplv.20006.kha)
- Kuparinen, Olli & Peltonen, Jaakko & Mustanoja, Liisa & Leino, Unni & Santaharju, Jenni. 2021. Lects in Helsinki Finnish - a probabilistic component modeling approach. *Language Variation and Change* 33(1). 1–26. (doi:10.1017/S0954394521000041)
- Labov, William. 1993. The unobservability of structure and its linguistic consequences. University of Ottawa.
- Labov, William. 2015. The discovery of the unexpected. *Asia-Pacific Language Variation* 1(1). 7–22.
- Lai, Li-Fang & Gooden, Shelome. 2024. Sociophonetic variation in the alveolar lateral in Yami. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 9(2). 195–238. (doi:10.1075/aplv.20003.lai)
- Langlois, Annie. 2006. Wordplay in teenage Pitjantjatjara. *Australian Journal of Linguistics* 26(2). 181–192.
- Lastra, Yolanda. 2009. Towards a study of language variation and change in Jonaz Chichimec. In Stanford, James N. & Preston, Dennis Richard (eds.), *Variation in Indigenous Minority Languages*, 153–172. Amsterdam: John Benjamins.
- Lavandera, Beatriz R. 1978. Where does the sociolinguistic variable stop? *Language in Society*. Cambridge University Press 7(2). 171–182. (doi:10.1017/S0047404500005510)
- Leonard, Jean Leo. 2009. A sociolinguistic sketch of vowel shifts in Kaqchikel: ATR-RTR parameters and redundancy markedness of syllabic nuclei in an Eastern Mayan language. In Stanford, James N. & Preston, Dennis Richard (eds.), *Variation in Indigenous Minority Languages*, 173–210. Amsterdam: Benjamins.
- Levon, Erez. 2012. Gender, prescriptivism, and language change: Morphological variation in Hebrew animate reference. *Language Variation and Change* 24(1). 33–58. (doi:10.1017/S095439451200004X)
- Lindsey, Kate L. 2021. Ende oration and final /n/-realisation. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 7(1). 30–61. (doi:10.1075/aplv.19013.lin)
- Lucas, Ceil & Bayley, Robert & Rose, Mary & Wulf, Alyssa. 2002. Location Variation in American Sign Language. *Sign Language Studies*. Gallaudet University Press 2(4). 407–440.
- Lüpke, Friederike. 2016. Uncovering small-scale multilingualism. *Critical Multilingualism Studies* 4(2). 35–74.
- Mansfield, John & Green, Ian. 2021. Fricative contrasts and neutralization in Marri Tjevin. *Australian Journal of Linguistics*. Routledge 41(2). 220–261. (doi:10.1080/07268602.2021.1957774)

- Mansfield, John & Leslie-O'Neill, Henry & Li, Haoyi. 2023. Dialect differences and linguistic divergence: A crosslinguistic survey of grammatical variation. *Language Dynamics and Change*. Brill 13(2). 232–276. (doi:10.1163/22105832-bja10026)
- Mansfield, John Basil. 2015a. Morphotactic variation, prosodic domains and the changing structure of the Murrinhpatha verb. *Asia-Pacific Language Variation* 1(2). 162–188.
- Mansfield, John Basil. 2015b. Consonant lenition as a sociophonetic variable in Murrinh Patha (Australia). *Language Variation and Change* 27(2). 203–225.
- Mansfield, John Basil & Saldana, Carmen & Hurst, Peter & Nordlinger, Rachel & Stoll, Sabine & Bickel, Balthasar & Perfors, Andrew. 2022. Category clustering and morphological learning. *Cognitive Science* 46(2). e13107. (doi:10.1111/cogs.13107)
- Marley, Alexandra. 2021. “I Speak My Language My Way!”—Young People’s Kunwok. *Languages*. Multidisciplinary Digital Publishing Institute 6(2). 88. (doi:10.3390/languages6020088)
- Meyerhoff, Miriam. 2009. Replication, transfer, and calquing: Using variation as a tool in the study of language contact. *Language Variation and Change* 21(3). 297–317. (doi:10.1017/S0954394509990196)
- Meyerhoff, Miriam. 2015. Turning variation on its head: Analysing subject prefixes in Nkep (Vanuatu) for language documentation. *Asia-Pacific Language Variation* 1(1). 78–108.
- Misnadin. 2021. Voice onset time and vowel quality in Madurese: The role of gender and dialect. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 7(2). 168–184. (doi:10.1075/aplv.20008.mis)
- Mudd, Katie & Lutzenberger, Hannah & Vos, Connie de & Fikkert, Paula & Crasborn, Onno & Boer, Bart de. 2020. The effect of sociolinguistic factors on variation in the Kata Kolok lexicon. *Asia-Pacific Language Variation*. John Benjamins 6(1). 53–88. (doi:10.1075/aplv.19009.mud)
- Nagy, Noami. 2009. The challenges of less commonly studied languages: Writing a sociogrammar of Faetar. In Stanford, James N. & Preston, Dennis Richard (eds.), *Variation in Indigenous Minority Languages*, 397–417. Amsterdam: Benjamins.
- Nettle, Daniel. 1999a. Is the rate of linguistic change constant? *Lingua* 108(2). 119–136. (doi:10.1016/S0024-3841(98)00047-3)
- Nettle, Daniel. 1999b. Using Social Impact Theory to simulate language change. *Lingua* 108(2). 95–117. (doi:10.1016/S0024-3841(98)00046-1)
- Nichols, Johanna. 1992. *Linguistic diversity in space and time*. Chicago: University of Chicago Press.
- Nichols, Johanna. 1995. Diachronically Stable Structural Features. In Andersen, Henning (ed.), *Historical Linguistics 1993: Selected Papers from the 11th International Conference on Historical Linguistics, Los Angeles*, 337–355. Amsterdam: Benjamins.
- Nikolaeva, Irina. 2014. *A grammar of Tundra Nenets. A Grammar of Tundra Nenets*. De Gruyter Mouton.
- Noglo, Kossi. 2009. Sociophonetic variation in urban Ewe. In Stanford, James N. & Preston, Dennis Richard (eds.), *Variation in Indigenous Minority Languages*, 229–249. Amsterdam: John Benjamins.
- Opara, U. Chiaka & Mbagwu, D. Ugochukwu. 2021. Linguistic variation in Emekuku-Igbo. *PREORC Journal of Arts and Humanities* 6(0).
- O’Shannessy, Carmel. 2008. Children’s production of their heritage language and a new mixed language. In Simpson, Jane & Wigglesworth, Gillian (eds.), *Children’s Language and Multilingualism*, 261–282.
- Owens, Jonathan & Dodsworth, Robin & Kohn, Mary. 2013. Subject expression and discourse embeddedness in Emirati Arabic. *Language Variation and Change* 25(3). 255–285. (doi:10.1017/S0954394513000173)
- Palakurthy, Kayla. 2019. *Many Ways to Sound Diné: Linguistic Variation in Navajo*. UC Santa Barbara. (<https://escholarship.org/uc/item/2vq427q3>) (Accessed January 21, 2025.)
- Palfreyman, Nick. 2020. Social meanings of linguistic variation in BISINDO (Indonesian Sign Language). *Asia-Pacific Language Variation*. John Benjamins Publishing Company 6(1). 89–118. (doi:10.1075/aplv.00008.pal)
- Povilonis, Natalie & Guy, Gregory. 2022. Authenticity in language ideology: Social variation in Chanka Quechua. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 8(2). 240–273. (doi:10.1075/aplv.22004.pov)
- Reed, Lauren W. 2020. “Switching caps”: Two ways of communicating in sign in the Port Moresby deaf community, Papua New Guinea. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 6(1). 13–52. (doi:10.1075/aplv.19010.ree)
- Ridge, Eleanor. 2022. Patterns of variation in subject-indexing prefixes in Vatlongos, Southeast Ambrym. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 8(1). 72–105. (doi:10.1075/aplv.19006.rid)
- Riverin-Coutlée, Josiane & Kapia, Enkeleida & Gubian, Michele. 2024. Dialect change and language attitudes in Albania. *Language Variation and Change* 36(2). 219–242. (doi:10.1017/S0954394524000103)
- Roberts, Gareth & Sneller, Betsy. 2020. Empirical foundations for an integrated study of language evolution. *Language Dynamics and Change* 10(2).

- Rocío Cruz Gómez, Sandra. 2008. *La variación diafásica en el Tseltal de Oxchuch, Chiapas: Identificación y análisis de variables sociolingüísticas*. Mexico, D.F.: CIESAS. (Masters thesis.)
- Romero, Sergio. 2009. Phonological markedness, regional identity, and sex in Mayan: the fricativization of intervocalic /l/ in K'iche'. In Stanford, James N. & Preston, Dennis Richard (eds.), *Variation in Indigenous Minority Languages*, 281–297. Amsterdam: John Benjamins.
- Romero, Sergio. 2015. The emergence of negative concord in Santa María Chiquimula K'ichee' (Mayan): A variationist perspective. *Language Variation and Change* 27(2). 187–201. (doi:10.1017/S0954394515000058)
- Royal, Anne Marie. 1985. *Male/female Pharyngealization Patterns in Cairo Arabic: A Sociolinguistic Study of Two Neighborhoods*. University of Texas at Austin. (PhD thesis.)
- Saad, George. 2020. *Variation and change in Abui: The impact of Alor Malay on an indigenous language of Indonesia*. Leiden University. (PhD Thesis.)
- Saldana, Carmen & Oseki, Yohei & Culbertson, Jennifer. 2021. Cross-linguistic patterns of morpheme order reflect cognitive biases: An experimental study of case and number morphology. *Journal of Memory and Language* 118. 104204. (doi:10.1016/j.jml.2020.104204)
- Satyanath, Shobha & Laskar, Nazrin B. 2009. Ethnicity, bilingualism, and variable clitic marking in Bishnupriya Manipuri. In Stanford, James N. & Preston, Dennis Richard (eds.), *Variation in Indigenous Minority Languages*, 441–461. Amsterdam: John Benjamins.
- Schembri, Adam & McKee, David & McKee, Rachel & Pivac, Sara & Johnston, Trevor & Goswell, Della. 2009. Phonological variation and change in Australian and New Zealand Sign Languages: The location variable. *Language Variation and Change* 21(2). 193–231. (doi:10.1017/S0954394509990081)
- Schnell, Stefan & Barth, Danielle. 2018. Discourse motivations for pronominal and zero objects across registers in Vera'a. *Language Variation and Change* 30(1). 51–81. (doi:10.1017/S0954394518000054)
- Schnell, Stefan & Barth, Danielle. 2020. Expression of anaphoric subjects in Vera'a: Functional and structural factors in the choice between pronoun and zero. *Language Variation and Change* 32(3). 267–291. (doi:10.1017/S0954394520000125)
- Schokkin, Dineke. 2021. Variable realisation of verb-final /n/ in Idi. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 7(1). 1–29. (doi:10.1075/aplv.19011.sch)
- Schwenter, Scott A. & Torres Cacoulios, Rena. 2010. Grammaticalization paths as variable contexts in weak complementarity in Spanish. In Walker, James A. (ed.), *Aspect in Grammatical Variation*, 13–26. Amsterdam: John Benjamins Publishing Company.
- Shain, Cory & Tonhauser, Judith. 2010. The synchrony and diachrony of differential object marking in Paraguayan Guaraní. *Language Variation and Change* 22(3). 321–346. (doi:10.1017/S0954394510000153)
- Sinnemäki, Kaius. 2014. Cognitive processing, language typology, and variation. *Wiley Interdisciplinary Reviews: Cognitive Science* 5(4). 477–487. (doi:10.1002/wcs.1294)
- Skilton, Amalia. 2017. Three speakers, four dialects: Documenting variation in an endangered Amazonian language. In Hildebrandt, Kristine A. & Jany, Carmen & Silva, Wilson (eds.), *Documenting Variation in Endangered Languages* (Language Documentation & Conservation Special Publication 13), 94–115. Honolulu: University of Hawai'i Press.
- Smith, Ian & Johnson, Steve. 2000. Kugu Nganhcara. In Dixon, R.M.W. & Blake, Barry J. (eds.), *The handbook of Australian languages*, vol. 5, 355–490. Melbourne: Oxford University Press.
- Sorensen, Arthur P. 1967. Multilingualism in the Northwest Amazon. *American Anthropologist*. [American Anthropological Association, Wiley] 69(6). 670–684.
- Stanford, James N. 2009. "Eating the food of our place": Sociolinguistic loyalties in multidialectal Sui villages. *Language in Society* 38. 287–309.
- Stanford, James N. 2016. A call for more diverse sources of data: Variationist approaches in non-English contexts. *Journal of Sociolinguistics* 20(4). 525–541.
- Stanford, James N. & Pan, Yanhong. 2013. The sociolinguistics of exogamy: Dialect acquisition in a Zhuang village. *Journal of SocioLinguistics* 17(5). 1–35.
- Strong, Katherine & Lindsey, Kate L. & Drager, Katie. 2022. Kawa and the variable stopping of obstruents in Ende. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 8(2). 150–173. (doi:10.1075/aplv.21003.str)
- Suokhrie, Kelhouvinuo. 2016. Clans and clanlectal contact: Variation and change in Angami. *Asia-Pacific Language Variation* 2(2). 188–214.
- Travis, Catherine E. & Ghina, Inas. 2021. Gender, mobility and contact: Stability and change in an Acehnese dialect. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 7(2). 142–167. (doi:10.1075/aplv.20007.tra)
- Trudgill, Peter. 2011. *Sociolinguistic typology: social determinants of linguistic complexity*. Oxford: Oxford University Press.

- Trudgill, Peter. 2017. Sociolinguistic Typology: Social Structure and Linguistic Complexity. In Aikhenvald, Alexandra Y. & Dixon, R. M. W. (eds.), *The Cambridge Handbook of Linguistic Typology* (Cambridge Handbooks in Language and Linguistics), 124–150. Cambridge: Cambridge University Press. (doi:10.1017/9781316135716.005)
- Vicente Ferre, Angel. 2017. *Formas nuevas para una lengua añeja: Análisis de la variación morfofonológica en el Náhuatl de San Miguel Canoa, La Resurrección y San Andrés Azumiatla, Puebla*. Mexico, D.F.: CIESAS. (PhD Thesis.)
- Wei, Monica X. & Sze, Felix Y. B. & Wong, Aaron Y. L. 2018. Gender variation in signs of sexual behaviour in Hong Kong Sign Language. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 4(1). 1–35. (doi:10.1075/aplv.17002.wei)
- Weinreich, Uriel & Labov, William & Herzog, Marvin. 1968. Empirical foundations for a theory of language change. In Lehmann, W. & Malkiel, Y. (eds.), *Directions for historical linguistics*, 95–198. Austin: University of Texas Press.
- Wilkinson, Melanie. 1991. *Djambarrpuyŋu: a Yolŋu variety of northern Australia*. University of Sydney. (PhD thesis.)
- Wilmoth, Sasha. 2022. *The Dynamics of Contemporary Pitjantjatjara: An Intergenerational Study*. University of Melbourne. (PhD thesis.)
- Wilmoth, Sasha & Defina, Rebecca & Loakes, Debbie. 2021. They talk mutumuṯu: Variable elision of tense suffixes in contemporary Pitjantjatjara. *Languages*. Multidisciplinary Digital Publishing Institute 6(2). 69. (doi:10.3390/languages6020069)
- Yang, Cathryn & Stanford, James N. & Luo, Chunxia & Zhang, Naluo. 2022. Generational differences in the low tones of Black Lahu. *Linguistics Vanguard*. De Gruyter Mouton 8(s5). 759–770. (doi:10.1515/lingvan-2021-0099)
- Yohana, Rafiki. 2009. *A Sociolinguistic Analysis of Variation in a Rural African Community*. University of Cape Town. (<http://hdl.handle.net/11427/3599>) (Accessed February 20, 2025.)
- Yu, Alan C. L. 2008. The phonetics of quantity alternation in Washo. *Journal of Phonetics* (Phonetic Studies of North American Indigenous Languages) 36(3). 508–520. (doi:10.1016/j.wocn.2007.10.004)
- Zen, Evynurul Laily & Starr, Rebecca Lurie. 2021. Variation and contact-induced change in Javanese phonology among multilingual children in Indonesia. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 7(2). 95–119. (doi:10.1075/aplv.20005.zen)
- Zhang, Jingwei. 2019. Tone mergers in Cantonese: Evidence from Hong Kong, Macao, and Zhuhai. *Asia-Pacific Language Variation*. John Benjamins Publishing Company 5(1). 28–49. (doi:10.1075/aplv.18007.zha)
- Zhang, Qing. 2005. A Chinese yuppie in Beijing: Phonological variation and the construction of a new professional identity. *Language in Society* 34(3). 431–466.